EXPERIMENT EVIDENCE / 2026

Quantum vs. classical optimization evidence

An evidence-led comparison of the saved BloodFlow-Q Experiment Reports. It reports what the experiments measured, including where classical methods lead.

CONCLUSION: NO QUANTUM SUPREMACY OR ADVANTAGE DEMONSTRATED

All QAOA evidence here comes from local classical simulation on synthetic, small instances. Measured solver runtimes are far slower than the classical baselines; one warm-start study improves feasible-shot sampling versus standard QAOA, which is an algorithmic result, not a quantum speedup.

FULL REVIEW RECORDS

1,410

1,377 completed; 33 skipped

PRIMARY SCENARIOS

15

75 runs per core method

PAIRED WARM-START RUNS

120

6 scenarios × 10 paired seeds

Measured mean runtime · October 24 review

All methods used the same 15-scenario primary family. Runtime includes optimization and simulator work for QAOA; the y-axis uses logarithmic bar scaling so the classical timings remain visible.

Interpretation: in this test setup, greedy averaged about 0.055 ms and warm QAOA about 562 ms. These are machine- and implementation-specific measurements, not a hardware comparison or a scaling law.

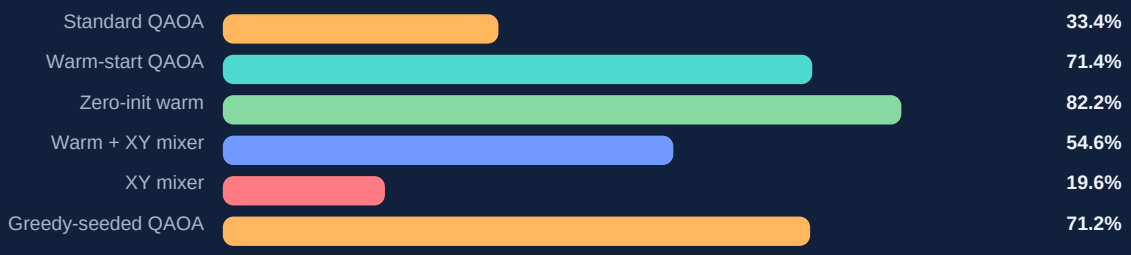
STUDY A · OCTOBER 24, 2026

What the full review measured

Primary-family methods used 15 small synthetic scenarios and five primary seeds (75 method-runs per method). Ablations cover only three scenarios and are not pooled into the comparison.

Feasible-shot probability · primary methods

Share of measured QAOA shots that decode to operationally feasible allocations. Classical solvers return a candidate rather than a shot distribution, so they are not plotted on this metric.



On these saved runs, zero-initialized warm QAOA had the highest mean feasible-shot rate (82.2%) among the listed variants; standard QAOA measured 33.4%. This measures feasible samples, not solution speed or advantage over classical optimization.

How a measured candidate is evaluated

- **Synthetic scenario**
- **Allocation objective + constraints**
- **QUBO → Ising cost operator**
- **QAOA on local simulator**
- **Measured bitstrings / shot counts**
- **Decode + classical feasibility validation**
- **Compare against greedy / exact / LP / MILP**

This hybrid workflow genuinely runs the QUBO/QAOA pipeline, but the quantum circuit is simulated with classical hardware in these experiments. “Measured” means simulator measurement samples, not quantum-device measurements.

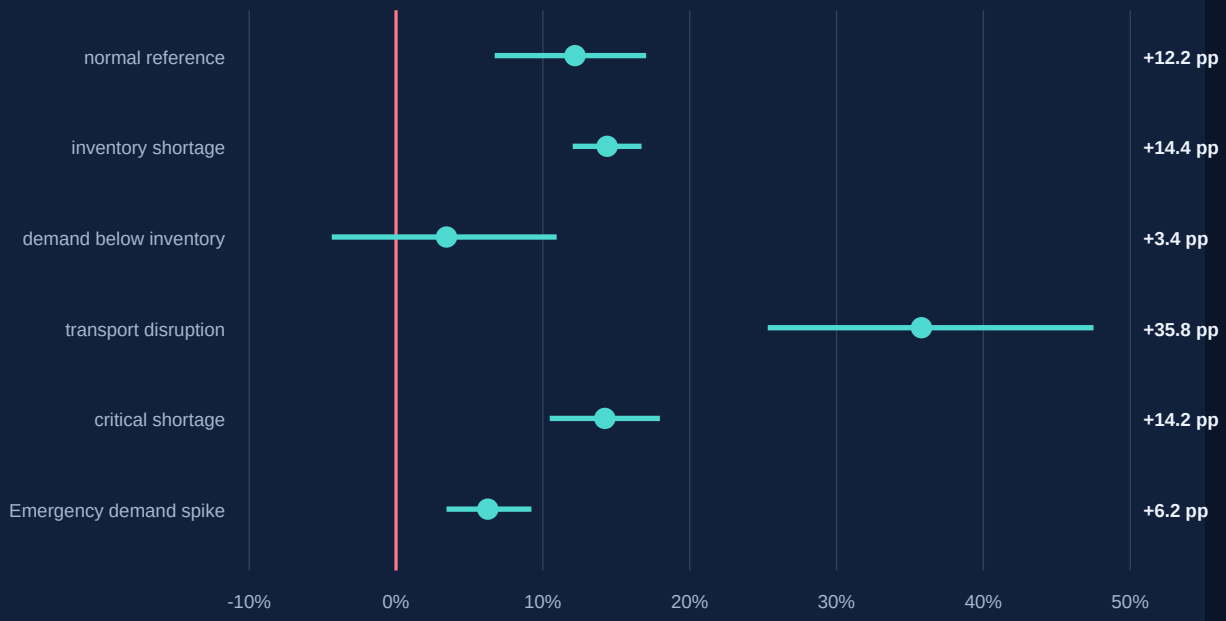
STUDY B · OCTOBER 22, 2026

A narrow QAOA improvement

Six synthetic scenarios; ten paired seeds each; p=1; 64 shots; COBYLA; eight iterations; one restart. The comparison is warm-start QAOA versus standard QAOA under this fixed protocol.

Paired change in feasible-shot rate by scenario

Point shows the mean paired difference (warm-start minus standard); whiskers are the saved 95% percentile bootstrap interval. Positive values favor warm-start sampling for this metric only.



STANDARD MEAN FEASIBLE SHOTS

17.7%

679 / 3,840 feasible shots

LP WARM-START MEAN

32.1%

1,231 / 3,840 feasible shots

PAIRED MEAN CHANGE

+14.4 pp

scenario-level CIs shown above

*The published report provides scenario-level bootstrap intervals; one scenario (demand below inventory) has an interval crossing zero. The average is across only six cases, and no broad population claim is justified.

Important classical contribution

The LP warm start is computed classically. It often supplies a strong starting solution and can itself reach exact references in this small study (60/60 warm-start runs recorded an exact optimum, versus 45/60 standard runs). This evidence supports a tested initialization strategy; it does not isolate a quantum contribution or establish superiority to classical solvers.

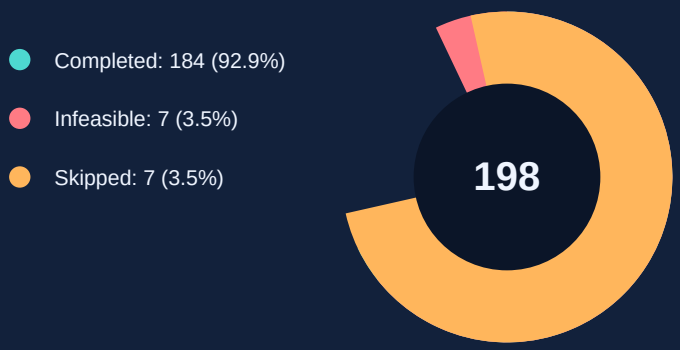
STUDY C · OCTOBER 15, 2026

Keep the older protocol separate

This legacy confirmatory benchmark has a different protocol and denominator from the newer studies. Its outcomes are shown as a separate historical evidence set, not pooled with the later full review or paired warm-start study.

198 recorded attempts · recorded statuses

The status pie accounts for all 198 rows in this report: 184 completed, 7 infeasible, and 7 skipped because the QAOA seed was infeasible.



Exact-reference matches · legacy study

Counts are shown with each method's actual denominator. Hybrid results exclude seven skipped cases and report exact matches among 53 evaluated feasible rows.

Classical multistart matched the exact reference in 60/60 attempts. Raw QAOA matched it in 29/53 feasible rows, with 7 infeasible rows. Hybrid local search matched 52/53 evaluated feasible rows, but the saved report says the preregistered improvement criterion was not met. This is not evidence of a QAOA win.

Measured mean runtime: classical multistart 0.747 ms; raw QAOA 139 ms; hybrid local search 139 ms. These are local measurements on the saved setup. Raw QAOA's mean objective for feasible results was 128.78; the hybrid mean was 90.57; classical multistart mean was 92.43. Objective averages are meaningful only in this legacy scenario set and protocol.

WHAT THE EVIDENCE CAN SUPPORT

Results, limitations & responsible wording

A compact reading guide for a demo or technical review.

What favored classical methods

For the full review, classical mean runtime was far lower: greedy 0.000055 s, exact 0.000268 s, LP 0.000771 s, MILP 0.00276 s, versus 0.458 s standard QAOA and 0.562 s warm QAOA. Classical solvers also provide direct feasible solutions without sampling.

What improved within QAOA

On the paired study, LP warm-start QAOA increased the measured feasible-shot rate by 14.4 percentage points on average relative to standard QAOA, under one small fixed protocol. In the full review, zero-initialized warm variants also had higher feasible-shot rates than standard QAOA. These are within-QAOA sampling findings.

What remains unknown

No quantum hardware was used; the benchmarks are small synthetic instances; some studies use different case sets; one warm-start study includes classical LP preprocessing; runtimes depend on this machine and implementation; the research protocol does not establish scaling, operational value, or a quantum contribution beyond classical preprocessing.

DO THESE RESULTS DEMONSTRATE QUANTUM ADVANTAGE? NO.

The current evidence does not show quantum supremacy, quantum advantage, faster optimization, clinical validity, or real hospital performance.

Recommended explanation

“Our prototype encodes a synthetic allocation problem as QUBO and runs QAOA on a local simulator. In a small paired experiment, an LP-informed warm start improved the fraction of feasible QAOA samples compared with standard QAOA. Classical methods still ran substantially faster in our saved tests, so we do not claim quantum advantage. The result is a reproducible research comparison, not a clinical or deployment validation.”

Method and source notes

Data were read from the saved Experiment Reports artifacts and summarized without rerunning or changing experiments. Studies are deliberately separated: October 24 full review (15 primary scenarios; 75 runs per core method), October 22 paired warm-start study (6 scenarios; 10 paired seeds; p=1; 64 shots), and October 15 legacy confirmatory study (198 attempts). Status counts and denominators are preserved. Runtime is wall-clock from the recorded local setup; it is not portable performance evidence.

Primary project references: docs/EXPERIMENT_RESULTS.md (October 24 review and October 15 legacy results); docs/EXPERIMENT_RESULTS_WARM_QAOA.md (October 22 paired warm-start study); frontend/public/full-review-study.json, frontend/public/warm-qaoa-paired-20261022.json, frontend/public/quantum-value-study.json (saved report data). Derived interactive chart data: frontend/public/comparative-evidence.json. Source SHA-256 values are embedded in that derived file.